Chapter 2: Financial Statements, Taxes, and Cash Flow

Multiple Choice · Chapters 2–7 · 53 questions · ★ = harder

1. A firm issues 200,000 new shares with a par value of $0.50 at $15 per share. How is the issue recorded on the balance sheet?

a. Common stock $3,000,000; paid-in surplus $0

b. Common stock $100,000; paid-in surplus $3,000,000

c. Common stock $100,000; paid-in surplus $2,900,000

d. Common stock $2,900,000; paid-in surplus $100,000

2. Consider these statements. I. IFRS is more principles-based than U.S. GAAP. II. Under IFRS (including K-IFRS) the statement of financial position usually lists non-current assets before current assets. III. U.S. GAAP is set by the IASB. Which are correct?

a. I only

b. I and II only

c. II and III only

d. I, II and III

3. Korea's national corporate tax rates are 10% up to ₩200 million, 20% from ₩200 million to ₩20 billion, 22% from ₩20 billion to ₩300 billion, and 25% above. A firm has taxable income of ₩21 billion. What are its average and marginal tax rates?

a. Average 22%; marginal 22%

b. Average 20%; marginal 20%

c. Average 20%; marginal 22%

d. Average 22%; marginal 20%

4. Which statement about corporate taxation is correct?

a. The U.S. has used a graduated corporate tax schedule since 2017.

b. OECD Pillar Two sets a 15% minimum effective tax for large multinationals; it is a floor, not the standard rate.

c. Most countries use a progressive corporate tax like Korea.

d. Pillar Two applies to every firm regardless of size.

5. Which statement about the U.S.-style balance sheet is correct?

a. Assets are listed in order of increasing liquidity.

b. Liabilities are listed in descending order of when they are due.

c. Assets are listed in decreasing order of liquidity, and liabilities in ascending order of when they are due.

d. It measures performance over a period of time.

Chapter 2: Financial Statements, Taxes, and Cash Flow — Answers

1. c
Step 1: Common stock = Shares × Par = 200,000 × 0.50 = $100,000
Step 2: Paid-in surplus = (Issue price − Par) × Shares = 14.50 × 200,000 = $2,900,000
Check: total paid-in capital = $3,000,000, the cash shareholders actually paid
Note: par is a nominal legal amount and economically arbitrary.

2. b
Statement I: correct
GAAP is more rules-based, IFRS more principles-based
Statement II: correct
GAAP lists liquid items first, IFRS lists non-current items first
Statement III: wrong
U.S. GAAP is set by the FASB under SEC oversight, IFRS by the IASB.

3. c
Step 1: Tax on first ₩200M = 0.10 × 200M = ₩20M
Step 2: Tax on next ₩19.8B = 0.20 × 19.8B = ₩3.96B
Step 3: Tax on last ₩1B = 0.22 × 1B = ₩0.22B
Step 4: Total tax = ₩4.2B
average = 4.2 / 21 = 20%
Marginal rate: the rate on the next won = 22%
Contrast: under the U.S. flat 21% rate, marginal = average.

4. b
U.S.: a flat 21% since the 2017 Tax Cuts and Jobs Act (previously graduated up to 35%)
Global: most countries use a single flat rate
a few, such as Korea, are progressive
Pillar Two: a 15% minimum effective tax for multinationals with revenue of €750M or more, a floor rather than a standard rate.

5. c
Assets: decreasing liquidity, cash first
Liabilities: ascending order of when due, current liabilities first
Balance sheet: a snapshot at a point in time
the income statement covers a period.

Chapter 3: Working with Financial Statements

Multiple Choice · Chapters 2–7 · 53 questions · ★ = harder

1. Firm X has a profit margin of 2%, total asset turnover of 4.0 and an equity multiplier of 2.5. Firm Y has a profit margin of 8%, total asset turnover of 1.0 and an equity multiplier of 2.5. Which statement is correct?

a. Firm X relies more on financial leverage than Firm Y.

b. Firm Y uses its assets more efficiently than Firm X.

c. Both earn an ROE of 20%: X through rapid asset turnover and Y through high margins, with equal leverage.

d. Firm Y earns the higher ROE because its margin is higher.

2. A firm has current assets of $708 (including cash of $98 and inventory of $422) and current liabilities of $548. What are its current, quick and cash ratios?

a. 1.29; 0.77; 0.18

b. 1.29; 0.52; 0.18

c. 0.77; 0.52; 0.18

d. 1.29; 0.52; 0.70

3. A firm has sales of $2,361, cost of goods sold of $1,344 and accounts payable of $344. Using the lecture's definition, what is the payables turnover and the average number of days taken to pay suppliers?

a. 6.86 times; 53.2 days

b. 3.91 times; 93.4 days

c. 3.91 times; 53.2 days

d. 6.86 times; 93.4 days

4. PepsiCo's current ratio is 0.85, while Coca-Cola's is 1.46. According to the lecture, why is PepsiCo's below-1 ratio not a red flag?

a. PepsiCo holds large cash reserves outside current assets.

b. Steady cash sales fund working capital, long supplier terms inflate payables, and strong credit lets it roll cheap short-term debt.

c. PepsiCo uses IFRS, which lowers the current ratio.

d. A current ratio below 1 is always healthier.

5. Coca-Cola has a debt-equity ratio of 2.26 and PepsiCo 4.26. Both earn about 40% ROE. Which statement is correct?

a. Coca-Cola's equity multiplier is 2.26 and PepsiCo's is 4.26.

b. Coca-Cola's equity multiplier is 3.26; PepsiCo reaches a similar ROE through faster turnover and more leverage.

c. Both firms reach their ROE mainly through profit margin.

d. PepsiCo has the higher profit margin.

6. Which of the following is a reason the lecture gives for why ratio comparisons can mislead?

a. Firms may time transactions around reporting dates (window dressing).

b. Ratios have a single correct value given by theory.

c. Peer groups cannot be formed from SIC or NAICS codes.

d. Time-trend analysis compares a firm only with competitors.

7. Using the lecture's definition, enterprise value = market value of the stock + book value of all liabilities − cash. A firm has market capitalization of $20,000, total liabilities of $2,150 (of which interest-bearing debt is $1,760), cash of $260 and EBITDA of $1,530. What is EV/EBITDA?

a. 14.05

b. 13.07

c. 14.31

d. 14.65

Chapter 3: Working with Financial Statements — Answers

1. c
Formula: ROE = Profit margin × Total asset turnover × Equity multiplier
Step 1: ROE(X) = 0.02 × 4.0 × 2.5 = 20%
Step 2: ROE(Y) = 0.08 × 1.0 × 2.5 = 20%
Step 3: Equity multipliers are equal (2.5), so leverage is the same
Reading: X turns assets over 4 times faster, Y earns a 4 times higher margin.

2. b
Current ratio = 708 / 548 = 1.29
Quick ratio = (708 − 422) / 548 = 0.52
Cash ratio = 98 / 548 = 0.18
Order: each ratio removes less liquid assets, so current > quick > cash.

3. b
Payables turnover = COGS / Accounts payable
Step 1: 1,344 / 344 = 3.91 times
Step 2: Days = 365 / 3.91 = 93.4 days
Watch for: payables turnover uses COGS, like inventory turnover
receivables turnover uses sales.

4. b
Reason 1: steady cash sales repay obligations as cash arrives
Reason 2: long payment terms to suppliers inflate current liabilities
Reason 3: strong credit lets it roll short-term debt at any time
Lesson: a large idle current-asset cushion would only lower returns.

5. b
EM = 1 + D/E
Coca-Cola: EM = 1 + 2.26 = 3.26
PepsiCo: EM = 1 + 4.26 = 5.26
Coca-Cola: high margin (27.3%) and low turnover (0.46)
PepsiCo: thin margin (8.8%), turnover 0.87 and higher leverage.

6. a
Cautions: no underlying theory, conglomerates without a clean peer group, global competitors with different standards, accounting method choices, different fiscal year-ends, capital structure, seasonality, window dressing
Benchmarks: time-trend (own history) and peer-group (SIC/NAICS codes).

7. c
Step 1: EV = 20,000 + 2,150 − 260 = 21,890
Step 2: EV/EBITDA = 21,890 / 1,530 = 14.31
Watch for: the lecture uses the book value of all liabilities, not only interest-bearing debt (which gives 14.05)
EBITDA = EBIT + depreciation and amortization.

Chapter 4: Introduction to Valuation: The Time Value of Money

Multiple Choice · Chapters 2–7 · 53 questions · ★ = harder

1. You invest $5,000 at 6% a year for 4 years. How much more do you have with compound interest than with simple interest?
   a. $1,312.38
   b. $1,200.00
   c. $112.38
   d. $0.00

2. $3,000 grows to $7,500 at 9% a year. Which statement is correct?
   a. It takes 2.5 times as long as $1,000 growing to $2,500.
   b. It takes 10.63 years, exactly as long as $1,000 growing to $2,500 at 9%.
   c. It takes 16.67 years, because 150% ÷ 9% = 16.67.
   d. The time cannot be found without the annuity formula.

3. An investment of $6,000 grows to $15,000 in 12 years. What annual rate did it earn?

a. 12.50%

b. 20.83%

c. 7.94%

d. 8.33%

Chapter 4: Introduction to Valuation: The Time Value of Money — Answers

1. c
Compound: $FV = 5{,}000 \times (1.06)^4 = 6{,}312.38$
Simple: $FV = 5{,}000 \times (1 + 0.06 \times 4) = 6{,}200.00$
Interest on interest = 6,312.38 − 6,200.00 = $112.38.

2. b
Formula: $t = \ln(FV / PV) / \ln(1 + r)$
Step 1: FV / PV = 7,500 / 3,000 = 2.5
Step 2: t = ln 2.5 / ln 1.09 = 10.63 years
Insight: only the ratio FV / PV matters, so $1,000 → $2,500 takes the same time.

3. c
Formula: $r = (FV / PV)^{1/t} - 1$
Step 1: $r = (15{,}000 / 6{,}000)^{1/12} - 1 = 7.94\%$
Watch for: 150% / 12 = 12.50% is simple interest
BA II Plus: 12 N, −6000 PV, 0 PMT, 15000 FV, CPT I/Y = 7.94.

Chapter 5: Discounted Cash Flow Valuation

Multiple Choice · Chapters 2–7 · 53 questions · ★ = harder

1. A lease requires $1,000 at the beginning of every month for 24 months. The rate is 6% compounded monthly. What is the lease worth today?

a. $22,562.87
b. $24,000.00
c. $22,675.68
d. $22,450.61

2. Lender A quotes 8.4% compounded monthly. Lender B quotes 8.5% compounded semiannually. Which loan is cheaper?

a. Lender A, because its quoted rate is lower
b. Lender A, because its EAR is 8.68% versus 8.73% for B
c. Lender B, because its EAR is 8.68% versus 8.73% for A
d. They cost the same

3. You save $400 at the end of every month in an account paying 6% compounded monthly. How many years until the balance reaches $30,000?

a. 63.85 years
b. 6.25 years
c. 5.32 years

4. You will receive $5,000 a year for 10 years, but the first payment arrives 4 years from today. The discount rate is 8%. What is the stream worth today?

a. $33,550.41
b. $24,660.55
c. $26,633.39
d. $28,764.07

5. You borrow $20,000 for 4 years with monthly payments. The bank quotes 8% compounded quarterly. What is the monthly payment?

a. $488.26
b. $416.67
c. $652.04
d. $487.76

6. You want to earn an effective annual rate of 10%, and the bank compounds monthly. What APR must you be quoted?

a. 10.00%
b. 10.47%
c. 9.57%
d. 0.83%

7. Consider these statements. I. Two loans with the same APR can cost different amounts. II. The gap between APR and EAR widens as compounding becomes more frequent. III. The gap narrows as the level of the rate rises. Which are correct?

a. I only
b. II and III only
c. I and II only
d. I, II and III

8. Which pairing of loan type and value is correct?

a. Pure discount loan: value = payment × annuity factor; used for mortgages
b. Interest-only loan: value = annuity of interest + PV of face; used for corporate bonds
c. Amortized loan: value = FV / $(1 + r)^t$; used for T-bills
d. Interest-only loan: principal is repaid in each payment

9. You borrow $8,000 for 5 years at 7%, repaid in equal annual payments of $1,951.13. How much interest is paid in year 2, and what is the balance after year 3?

a. $560.00; $3,527.67
b. $462.62; $3,200.00
c. $462.62; $3,527.67
d. $390.13; $3,527.67

10. A payment of $1,000 arrives one year from today and grows at 3% a year for a total of 10 payments. The discount rate is 8%. What is the present value?

a. $6,710.08
b. $20,000.00
c. $7,550.13
d. $10,000.00

Chapter 5: Discounted Cash Flow Valuation — Answers

1. c
Step 1: Monthly rate r = 6% / 12 = 0.5%
Step 2: PV (ordinary) = 1,000 × [1 − 1/(1.005)^{24}] / 0.005 = 22,562.87
Step 3: PV (due) = 22,562.87 × 1.005 = 22,675.68
BA II Plus: 24 N, 0.5 I/Y, −1000 PMT, 0 FV, CPT PV = 22,562.87
Then: × 1.005 = 22,675.68 (or BGN mode).

2. c
Step 1: EAR(A) = (1 + 0.084/12)^{12} − 1 = 8.73%
Step 2: EAR(B) = (1 + 0.085/2)^{2} − 1 = 8.68%
Step 3: Compare EARs, never quoted rates
Answer: B is cheaper (lower EAR).

3. c
Step 1: 30,000 = 400 × [(1.005)^{t} − 1] / 0.005
Step 2: (1.005)^{t} = 1.375
Step 3: t = ln 1.375 / ln 1.005 = 63.85 months
Step 4: Years = 63.85 / 12 = 5.32
BA II Plus: 0.5 I/Y, −400 PMT, 0 PV, 30000 FV, CPT N = 63.85 (months).

4. c
Step 1: The annuity formula gives the value one period before the first payment, at date 3
Step 2: PV_3 = 5,000 × [1 − 1/(1.08)^{10}] / 0.08 = 33,550.41
Step 3: PV_0 = 33,550.41 / (1.08)^{3} = 26,633.39
Watch for: discounting 4 years is the common error
BA II Plus: 10 N, 8 I/Y, −5000 PMT, 0 FV, CPT PV = 33,550.41
then 3 N, 0 PMT, −33550.41 FV, CPT PV = 26,633.39.

5. d
Step 1: Quarterly rate = 8% / 4 = 2%
Step 2: Effective monthly rate = (1.02)$^{1/3}$ − 1 = 0.6623%
Step 3: 20,000 = C × [1 − 1/(1.006623)^{48}] / 0.006623
Step 4: C = $487.76
Watch for: 8% / 12 assumes monthly compounding and gives $488.26
BA II Plus: 48 N, 0.6623 I/Y, 20000 PV, 0 FV, CPT PMT = −487.76.

6. c
Step 1: Monthly rate = (1.10)$^{1/12}$ − 1 = 0.7974%
Step 2: APR = 12 × 0.7974% = 9.57%
Check: (1 + 0.0957/12)^{12} − 1 = 10%
ICONV: EFF = 10, C/Y = 12, CPT NOM = 9.57.

7. c
Statement I: correct
APR is only a quoting convention
Statement II: correct
e.g. 15% APR gives 15.56% semiannually but 16.18% daily
Statement III: wrong
the gap widens with the level of the rate
Rule: compare only EARs.

8. b
Pure discount: one payment at maturity
value = FV / (1 + r)^{t}
T-bills and zero coupon bonds
Interest-only: interest each period, principal at the end
value = annuity of interest + PV of face
corporate bonds
Amortized: each payment covers interest and part of principal
mortgages and car loans.

9. c
Step 1: Year 1 interest = 0.07 × 8,000 = 560.00
principal = 1,951.13 − 560.00 = 1,391.13
Step 2: Year 2 interest = 0.07 × (8,000 − 1,391.13) = 462.62
Step 3: Balance after year 3 = PV of the 2 remaining payments = 1,951.13 × [1 − 1/(1.07)^{2}] / 0.07 = 3,527.67
Rule: the balance at any date is the PV of the payments still to be made.

10. c
Formula: PV = C × {1 − [(1 + g) / (1 + r)]^{t}} / (r − g)
Step 1: (1.03 / 1.08)^{10} = 0.6225
Step 2: PV = 1,000 × (1 − 0.6225) / 0.05 = 7,550.13
Contrast: a level annuity gives 6,710.08 and a growing perpetuity 20,000.

Chapter 6: Interest Rates and Bond Valuation

Multiple Choice · Chapters 2–7 · 53 questions · ★ = harder

1. A $1,000 bond pays an 8% annual coupon, matures in 3 years and now sells at par. If its yield rises to 9%, by how much does its price change?

a. −1.00%
b. −2.53%
c. −3.00%
d. +2.62%

2. Four bonds currently yield 8%. Which bond's price changes most when the market yield rises by one percentage point?

a. 5-year maturity, 10% coupon

b. 5-year maturity, 4% coupon

c. 20-year maturity, 10% coupon

d. 20-year maturity, 4% coupon

3. A bond's quoted yield to maturity equals the return an investor actually earns only under which conditions?

a. The bond is held to maturity, it does not default, and coupons are reinvested at the YTM.

b. The bond is sold before maturity at par.

c. Interest rates fall after purchase.

d. The coupon rate equals the current yield.

4. A $1,000 bond pays a 6% coupon semiannually, matures in 10 years and sells for $925. What is its yield to maturity (quoted annually)?

a. 3.53%

b. 6.49%

c. 7.06%

d. 7.18%

5. A bond with an 8% coupon rate and 8 years to maturity trades below par. Which ordering of its yields is correct?

a. YTM < current yield < coupon rate

b. Current yield < coupon rate < YTM

c. Coupon rate = current yield = YTM

d. Coupon rate < current yield < YTM

6. A municipal bond yields 4.2% and a comparable taxable bond yields 6.0%. Three investors face marginal tax rates of 20%, 30% and 35%. Which statement is correct?

a. All three prefer the taxable bond.

b. All three prefer the muni.

c. Only the 35% investor prefers the muni; the 30% investor is indifferent.

d. Only the 20% investor prefers the muni.

7. A $1,000 bond pays a 6% coupon semiannually and matures in 10 years. Investors require an effective annual return of 8.16%. What is the bond's price?

a. $1,000.00

b. $854.26

c. $864.10

d. $856.10

8. Consider these statements. I. Notes are unsecured issues, usually maturing in under ten years. II. A private placement is registered and sold to many investors. III. In registered form, the issuer records the owner and pays the coupon to that owner. Which are correct?

a. I only

b. I and III only

c. II and III only

d. I, II and III

9. Protective covenants restrict what management may do. Why do firms accept them?

a. Regulators require every bond to carry the same covenants.

b. The reduction in the yield the firm must pay is worth more than the flexibility it gives up.

c. Covenants raise the firm's credit rating to AAA.

d. Covenants are costless to the firm.

10. A bond rated BBB− by S&P is downgraded to BB+. Which statement is most accurate?

a. It remains investment grade.

b. It moves from the lowest investment-grade tier to speculative grade, which many institutional mandates forbid holding.

c. Its coupon rate is automatically raised.

d. It is now in default.

11. Which statement about U.S. Treasury securities is correct?

a. Treasury bills mature in two to ten years.

b. Treasury interest is exempt from federal income tax.

c. Bills mature in a year or less, notes in two to ten years, and bonds beyond that; interest is exempt from state and local tax.

d. Treasuries carry the same default risk as municipal bonds.

12. Why is the term structure of interest rates measured on pure discount, default-free bonds?

a. Because coupon bonds are not traded.

b. To remove default risk and coupon effects so that only time to maturity differs.

c. Because zero coupon bonds have no interest rate risk.

d. To include a liquidity premium.

13. Which statement is correct?

a. A floating-rate bond's coupon is fixed, but its maturity changes.

b. Income bonds are popular with issuers and investors alike.

c. A floating-rate bond's coupon resets against a reference rate and often has a collar (floor and ceiling).

d. A put bond lowers the bond's value.

Chapter 6: Interest Rates and Bond Valuation — Answers

1. b
Step 1: At 8% the price is $1,000 (coupon rate = YTM)
Step 2: At 9%: Price = $80 \times [1 - 1/(1.09)^3] / 0.09 + 1{,}000 / (1.09)^3 = 974.69$
Step 3: Change = 974.69 / 1,000 − 1 = −2.53%
Rule: prices and yields move in opposite directions.

2. d
Maturity effect: a longer bond has more cash flows exposed, and the distant face value swings most
Coupon effect: a lower coupon pushes more value to maturity, so it is more sensitive
Answer: the 20-year, 4% coupon bond combines both effects.

3. a
YTM: the single discount rate that sets the PV of the promised cash flows equal to the price (the bond's IRR)
Assumption 1: held to maturity
Assumption 2: no default (YTM is a promised yield)
Assumption 3: coupons reinvested at the YTM
If rates fall, reinvested coupons earn less and the realized return falls short.

4. c
Step 1: t = 20 periods
C = 30
Price = 925
Step 2: Solve $925 = 30 \times [1 - 1/(1 + r)^{20}] / r + 1{,}000 / (1 + r)^{20}$
Step 3: r = 3.53% per six months
Step 4: Quoted YTM = 3.53% × 2 = 7.06%
BA II Plus: 20 N, −925 PV, 30 PMT, 1000 FV, CPT I/Y = 3.53
Check: price below par, so YTM > coupon rate.

5. d
Coupon rate = Annual coupon / Face value
Current yield = Annual coupon / Price
it is higher because the price is below face
YTM: also counts the capital gain, so it is highest
Discount bond: coupon rate < current yield < YTM.

6. c
Formula: Equivalent taxable yield = 4.2% / (1 − t)
20%: 4.2 / 0.80 = 5.25% < 6.0%, taxable is better
30%: 4.2 / 0.70 = 6.00%, indifferent
35%: 4.2 / 0.65 = 6.46% > 6.0%, muni is better
Lesson: munis are held mainly by high-bracket investors.

7. c
Step 1: Semiannual rate = $(1.0816)^{1/2} - 1 = 4.00\%$
Step 2: t = 20
C = 30
Step 3: Price = $30 \times [1 - 1/(1.04)^{20}] / 0.04 + 1{,}000 / (1.04)^{20} = 864.10$
Watch for: 8.16% / 2 = 4.08% treats an effective rate as a quoted rate and gives $854.26
BA II Plus: 20 N, 4 I/Y, 30 PMT, 1000 FV, CPT PV = −864.10.

8. b
Statement I: correct
Statement II: wrong
a public issue is registered and sold to many investors, a private placement goes to a few institutions and skips registration
Statement III: correct
in bearer form whoever holds the certificate is paid, and bearer bonds are now rare.

9. b
Purpose: covenants stop the firm from expropriating its creditors
Cost: they restrict decisions management might want to take
Trade-off: firms accept them because the lower yield is worth more than the lost flexibility.

10. b
BBB− / Baa3: the lowest investment-grade rating
BB+ / Ba1: the top of speculative grade ("junk")
Consequence: many institutional mandates forbid holding anything below BBB−
Ratings assess the likelihood of payment and creditor protection, not default itself.

11. c
Maturities: bills ≤ 1 year
notes 2 to 10 years
bonds beyond 10 years
Tax: exempt from state and local income taxes, but not federal
Risk: no default risk, unlike municipal bonds.

12. b
Term structure: the relationship between time to maturity and yield, all else equal
Pure discount: removes coupon effects
Default-free: removes default risk premium
Components: real rate + inflation premium + interest rate risk premium.

13. c
Floater: coupon resets periodically against a reference rate, with a floor and a ceiling
Income bond: coupons paid only if the firm has sufficient income
investors read them as a signal of weakness, so they are rare
Put bond: the holder may force a buyback, which raises the bond's value.

Chapter 7: Equity Markets and Stock Valuation

Multiple Choice · Chapters 2–7 · 53 questions · ★ = harder

1. A company just paid a dividend of $2.00. Dividends grow at 5% a year forever and the required return is 12%. What will the stock be worth five years from now?

a. $36.47
b. $30.00
c. $38.29
d. $40.20

2. The required return on a stock is 9%. The next dividend is $1.00 and the growth rate is assumed to be 10% forever. What does the dividend growth model say?

a. The price is $100, so the stock is undervalued.
b. The price is $11.11 because growth is ignored when g exceeds R.
c. The price is −$100, which shows the assumption g > R cannot hold forever.

3. A stock sells for $50. It just paid a dividend of $2.00, and investors require 11%. What constant growth rate is the market implying?

a. 4.00%
b. 7.00%
c. 6.73%
d. 5.50%

4. A firm earns an ROE of 15% and pays out 40% of its earnings as dividends. Next year's earnings per share are expected to be $3.00, and investors require 13%. What is the stock worth?

a. $12.86
b. $20.00
c. $30.00
d. $13.33

5. Han River Utilities has preferred stock with a $100 par value and an annual dividend rate of 7.5%, paid quarterly. The dividend never changes. Investors require 9%, compounded monthly. What is the price of one preferred share?

a. $83.33
b. $82.71
c. $79.95
d. $250.00

6. Mirae Logistics paid a dividend of $1.20 four years ago, and the dividend just paid was $1.62. Dividends will keep growing at the same average annual rate. Investors require 14%. What is the price today?

a. $33.56
b. $26.09
c. $28.13

7. Bitgoeul Biotech just paid a dividend of $2.00. Dividends grow 25% next year, 18% the year after and 12% in year 3, then 6% forever. The required return is 14%. What is the price today?

a. $32.61

b. $34.57

c. $26.50

d. $36.24

8. Hanbit Robotics will pay no dividend for the next three years. It will pay $2.00 at the end of year 4 and $2.50 at the end of year 5, after which dividends grow at 4% forever. The required return is 15%. What is the price today?

a. $14.14

b. $21.50

c. $13.69

d. $12.29

9. Gyeongbok Software just paid a dividend of $1.50. Dividends grow at 15% a year for five years and at 5% forever after that. The required return is 16%. Using the two-stage model, what is the price today?

a. $20.37

b. $21.02

c. $14.32

d. $19.13

10. Cheongdam Mobility had sales of $480 million and a net loss of $36 million. It has 24 million shares and pays no dividend. Comparable firms have price-sales ratios of 2.1, 2.6, 1.9 and 2.4, and an average P/E of 22. Estimate the stock price.

a. −$33.00

b. $48.00

c. $45.00

d. $1,080.00

11. On the NYSE, which party executes customer orders on the trading floor?

a. The designated market maker

b. Floor brokers

c. Supplemental liquidity providers

d. Electronic communications networks

12. What was the main effect of electronic communications networks (ECNs) on NASDAQ?

a. They replaced market makers with a physical trading floor.

b. They let customers post their own bids and offers, narrowing spreads and blurring the line between dealer and auction markets.

c. They widened spreads by adding dealers.

d. They required all trades to be matched by a DMM.

13. Which statement about voting and classes of stock is correct?

a. One share, one vote is required by law in every country.

b. Dual-class shares are mainly used to pay higher dividends.

c. One share, one vote is a matter of corporate law and listing rules, and it varies by jurisdiction.

d. Proxy voting gives dissident shareholders an advantage over incumbents.

14. Which statement about preferred stock is correct?

a. Preferred holders rank ahead of creditors for their stated value.

b. Preferred holders rank ahead of common holders for a stated value, and most issues are cumulative.

c. Preferred stock always has a maturity date.

d. Common dividends may be paid while preferred dividends are unpaid.

15. A stock quote shows a last price of $48.00, an indicated annual dividend of $1.20 and a P/E of 16. What are the dividend yield and the earnings per share?

a. 2.50%; $768.00

b. 2.50%; $3.00

c. 40.00%; $3.00

d. 0.025%; $16.00

Chapter 7: Equity Markets and Stock Valuation — Answers

1. c
Rule: the model works from any date, using the dividend that follows
Step 1: $D_6 = 2.00 \times (1.05)^6 = 2.68$
Step 2: $P_5 = D_6 / (R - g) = 2.68 / 0.07 = \$38.29$
Check: $P_0 = 2.10 / 0.07 = 30.00$ and $P_0 \times (1.05)^5 = 38.29$.

2. c
Step 1: $P_0 = D_1 / (R - g) = 1.00 / (0.09 - 0.10) = -\$100$
Step 2: A negative or infinite price means the assumption is impossible
Rule: the model requires R > g
no firm can outgrow its required return forever.

3. c
Setup: $P_0 = D_0 \times (1 + g) / (R - g)$
Step 1: $50 = 2.00 \times (1 + g) / (0.11 - g)$
Step 2: $5.50 - 50g = 2.00 + 2.00g$
Step 3: $52g = 3.50$, so $g = 6.73\%$
Check: $D_1 = 2.1346$ and $2.1346 / (0.11 - 0.0673) = 50.00$.

4. c
Step 1: Retention ratio $b = 1 - 0.40 = 0.60$
Step 2: $g = ROE \times b = 0.15 \times 0.60 = 9\%$
Step 3: $D_1 = 0.40 \times 3.00 = 1.20$
Step 4: $P_0 = D_1 / (R - g) = 1.20 / (0.13 - 0.09) = \$30.00$
Idea: growth comes from earnings the firm retains and reinvests at its ROE.

5. b
Step 1: Quarterly dividend $D = 0.075 \times 100 / 4 = \$1.875$
Step 2: Quarterly rate $R = (1 + 0.09/12)^3 - 1 = 2.2669\%$
Step 3: $P_0 = D / R = 1.875 / 0.022669 = \$82.71$
Watch for: $0.09 / 4 = 2.25\%$ ignores monthly compounding and gives \$83.33.

6. c
Step 1: $1.20 \times (1 + g)^4 = 1.62$, so $g = (1.35)^{1/4} - 1 = 7.79\%$
Step 2: $D_1 = 1.62 \times 1.0779 = 1.7462$
Step 3: $P_0 = 1.7462 / (0.14 - 0.0779) = \$28.13$
Watch for: $35\% / 4 = 8.75\%$ is a simple average and overstates g.

7. d
Step 1: $D_1 = 2.50$
$D_2 = 2.95$
$D_3 = 3.304$
Step 2: $P_3 = D_4 / (R - g) = 3.304 \times 1.06 / 0.08 = 43.778$
Step 3: $P_0 = 2.50 / 1.14 + 2.95 / (1.14)^2 + (3.304 + 43.778) / (1.14)^3 = \$36.24$
Watch for: P_3 uses D_4 and is discounted 3 years, not 4.

8. a
Step 1: $P_5 = D_6 / (R - g) = 2.50 \times 1.04 / 0.11 = 23.636$
Step 2: $P_0 = 2.00 / (1.15)^4 + (2.50 + 23.636) / (1.15)^5 = \$14.14$
Watch for: the three zero years must be counted (CF: F01 = 3), or every later cash flow is discounted too few periods.

9. b
Step 1: First stage $= [1.50 \times 1.15 / (0.16 - 0.15)] \times [1 - (1.15 / 1.16)^5] = 7.308$
Step 2: $D_5 = 1.50 \times (1.15)^5 = 3.0170$
Step 3: $P_5 = 3.0170 \times 1.05 / 0.11 = 28.799$
Step 4: $P_0 = 7.308 + 28.799 / (1.16)^5 = 7.308 + 13.712 = \$21.02$
Watch for: $R - g_1 = 1\%$ is tiny, so keep full precision.

10. c
Step 1: Sales per share $= 480 / 24 = \$20.00$
Step 2: Benchmark P/S $= (2.1 + 2.6 + 1.9 + 2.4) / 4 = 2.25$
Step 3: $P_0 = 2.25 \times 20.00 = \$45.00$
Why not P/E: EPS $= -36 / 24 = -\$1.50$, so $22 \times (-1.50)$ gives a meaningless negative price.

11. b
DMM: posts continuous two-sided quotes and provides liquidity when natural buyers or sellers are absent
Floor brokers: execute customer orders on the floor
their numbers have fallen sharply
SLMs: electronic members adding quotes with incentives, without the DMM's obligations.

12. b
ECNs: customers post bids and offers directly, competing with dealers
Effect 1: spreads narrowed substantially
Effect 2: dealer and auction markets converged
NYSE became largely electronic and NASDAQ acquired auction-like features.

13. c
Dual-class: classes differ in voting power, letting founders raise capital while keeping control
One share, one vote: set by corporate law and listing rules and varies by jurisdiction
Proxy: management routinely solicits proxies, so incumbents start with the advantage.

14. b
Stated value: preferred ranks ahead of common for a stated (par) value, but behind every creditor
Cumulative: unpaid dividends accumulate as arrearages and must be cleared first
most issues are cumulative
Maturity: many issues have none.

15. b
Yield = Dividend / Price $= 1.20 / 48.00 = 2.50\%$
P/E = Price / EPS, so EPS $= 48.00 / 16 = \$3.00$.